Why AI Text Detectors Struggle with Banglish

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ami kal class e jabo na, too tired. keu notes ta share korte parba?

Read that message again and ask yourself: did a person write it, or did an AI?

If you grew up in Bangladesh, the message looks completely ordinary. It's Banglish: Bangla written in the Roman alphabet, mixed freely with English. We use it in Messenger chats, Facebook comments, product reviews and class group chats every day. More and more Banglish text is now produced by large language models (LLMs) like ChatGPT and Gemini, and it can be very hard to tell apart from ours.

This matters because AI text detectors are already being used to judge real people's writing. Teachers check assignments with them, and platforms use them to filter spam and fake reviews. Almost all of these detectors were built for standard English. In this post I explain how they work and why Banglish breaks their core assumptions. Then I test three of them on 200 Banglish product reviews. The short version: **none of them did better than a coin flip, and one flagged 71 out of 100 real human reviews as AI-written.**

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1. How AI text detectors decide

Most detectors fall into one of three families.

Statistical detectors

A language model assigns a probability to every word it reads. Text it finds predictable has low **perplexity**, and text that surprises it has high perplexity. AI-generated text tends to be "smooth": it usually picks likely words, so a language model finds it unsurprising. Human writing is burstier. We use odd word choices, typos, slang and sudden topic shifts.

Early tools like **GLTR** (Gehrmann et al., 2019) visualized this by coloring each word by how highly a model ranked it. Later methods such as **DetectGPT** (Mitchell et al., 2023) and **Binoculars** (Hans et al., 2024) refined the idea into zero-shot detectors, which need no training on labelled examples.

Trained classifiers

The second approach is plain supervised learning. You collect thousands of human-written and AI-generated texts, then fine-tune a model such as RoBERTa to tell them apart. These classifiers can be very accurate, but only on the kind of text they were trained on.

Watermarking

The third approach puts the job on the generator. While producing text, the model quietly favors certain words in a pattern that a matching detector can later check. Watermarking can work well, but only for models that choose to use it. It can't tell you anything about text from a model that doesn't.

What they have in common

Each of these methods depends on some idea of what "normal" text looks like, whether that's a language model's perplexity, a classifier's training data, or a pattern in the vocabulary. In practice, "normal" has mostly been learned from standard English.

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2. What makes Banglish different

Code-switching

Banglish speakers switch languages mid-sentence, sometimes in the middle of a phrase: *"meeting ta postpone hoye geche, so kal free achi."* To a model trained on English, the Bangla words look like noise, and to a model trained on Bangla script they look like a foreign alphabet. This kind of text is common in everyday writing, but it's rare in training data.

No fixed spelling

Banglish has no standard spelling. The word for "good" appears as *valo*, *bhalo*, *vlo* or *vhalo*. All of these appear in the real reviews I used. One writer can spell the same word two ways in one message. To a statistical detector, this variation looks like noise.

Tokenization

Before a model reads any text, a **tokenizer** splits it into pieces. I compared one sentence written three ways: in English, in Bangla script and in Banglish.

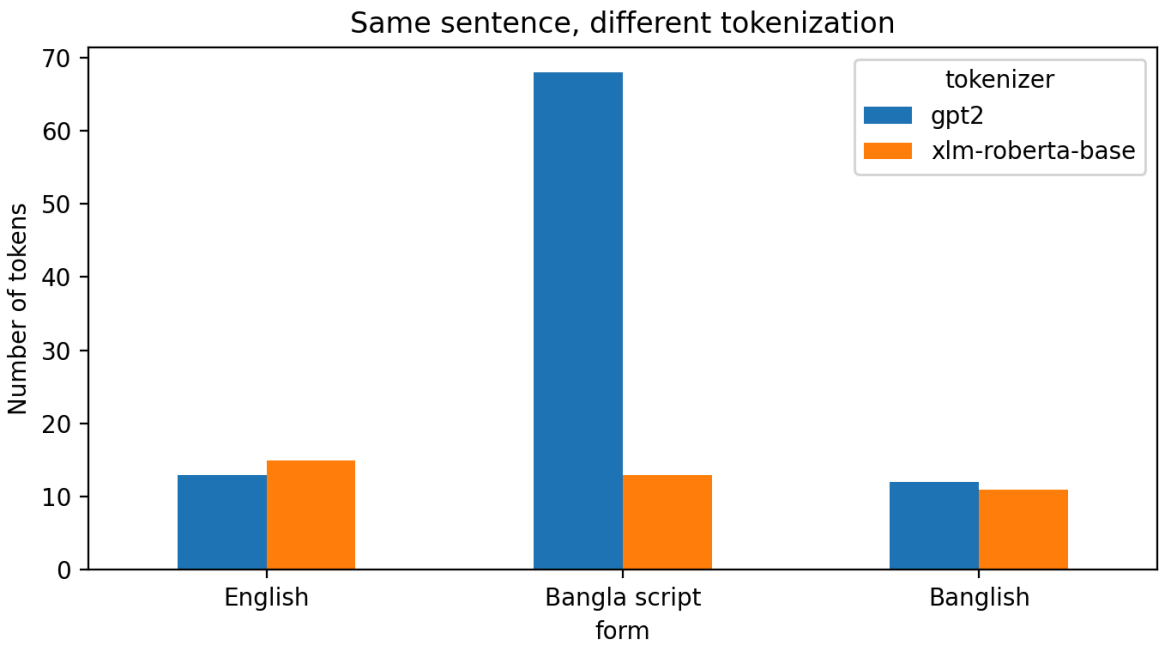


Figure 1. Number of tokens for the same sentence in English, Bangla script and Banglish, using GPT-2's tokenizer and XLM-R's tokenizer.

	GPT-2 (English-focused)	**XLM-R (multilingual)**
English	13 tokens	15 tokens
Bangla script	**68 tokens**	13 tokens
Banglish	12 tokens	11 tokens

Two things stand out. First, GPT-2 needed **68 tokens** for the Bangla-script sentence, more than five times as many as for English, because it breaks each Bangla character into raw bytes. XLM-R, which was trained on many languages, needed only 13.

Second, Banglish *looks* efficient at 12 tokens, but the token count hides the problem. GPT-2 split *kal* ("tomorrow") into k + `al` and *jabo* ("will go") into `jab` + o. It reads Banglish as if it were English, so it cuts words in meaningless places. XLM-R does a bit better with *kal*, but it still splits *jabo* into `ja` + bo. In both cases the model gets English-shaped pieces with no Bangla meaning attached. Probability scores built on those pieces, which perplexity-based detectors depend on, may not mean what the detector assumes.

(This is one sentence. It illustrates how tokenizers behave, and isn't a measurement across a dataset.)

Very little data

Detectors have rarely seen Banglish from humans or from AI. We already know detectors can be unfair to people who don't write "textbook" English. Liang et al. (2023) found that popular GPT

detectors often flagged essays by non-native English writers as AI-generated. Banglish writers could face a similar bias, and my experiment suggests they do.

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3. A small experiment

To see whether this happens in practice, I ran a small test.

Setup

- **Human text:** 100 real Banglish product reviews from an e-commerce review dataset [BanglaRev] I only used reviews posted **before 30 November 2022**, the day ChatGPT was released, so none of them could have been written by a modern chatbot.
- **AI text:** 100 Banglish product reviews, 50 from **Gemini** and 50 from **ChatGPT**, generated on similar shopping topics such as delivery, quality, price and refunds.
- **Detectors:**
    1. a RoBERTa classifier trained to detect ChatGPT text (`Hello-SimpleAI/chatgpt-detector-roberta`)
    2. OpenAI's older GPT-2 output detector (`openai-community/roberta-base-openai-detector`)
    3. a zero-shot perplexity score from BLOOM-560m, a small multilingual language model, where lower perplexity counts as "AI" (the threshold is the median)
- **Metrics:** accuracy, false-positive rate (human text flagged as AI), and false-negative rate (AI text passing as human)

The full code is in the Colab notebook:
https://colab.research.google.com/drive/1A_5cKP-HulXIOz3C2_SC3QrxM52M-dKa

Results

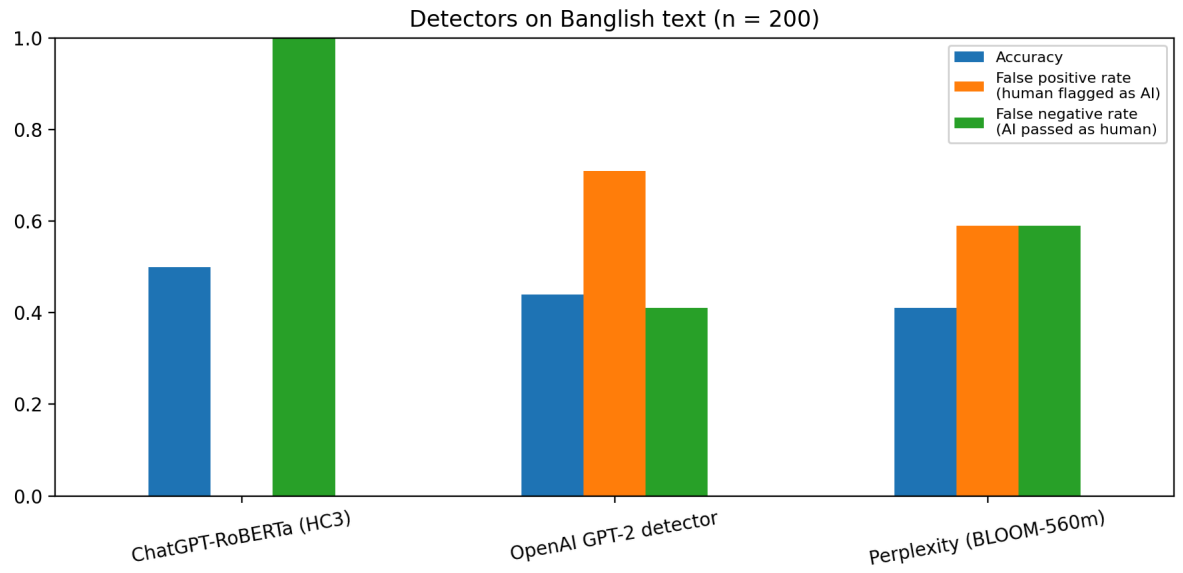


Figure 2. Accuracy, false-positive rate and false-negative rate of each detector on 200 Banglish reviews.

Detector	Accuracy	Human flagged as AI	AI passed as human
ChatGPT-RoBERTa (HC3)	50%	0%	**100%**
OpenAI GPT-2 detector	44%	**71%**	41%
Perplexity (BLOOM-560m)	41%	59%	59%

With equal numbers of human and AI texts, guessing at random would score about 50%. **None of the three detectors did better than that.** Each one failed in a different way.

The ChatGPT detector was blind. It labelled all 200 texts as human. The highest "AI" probability it gave any text was 2%, for a review that ChatGPT actually wrote. Its 50% accuracy only comes from the fact that half the dataset really was human. It could be accurate on English ChatGPT text, but on Banglish it never raised an alarm once.

The GPT-2 detector was the opposite: it accused real people. It flagged **71 of the 100 human reviews** as AI, 21 of them with more than 95% confidence. Meanwhile it let 41 of the 100 AI reviews through as human. In other words, a real Daraz customer was *more* likely to be flagged than a chatbot.

The perplexity detector got the theory backwards. The idea behind perplexity detection is that AI text is smoother, so it should score *lower* than human text. On Banglish the opposite happened:

Group	Median perplexity
Human reviews	3,655
Gemini	4,290
ChatGPT	10,930

Real human reviews were the *least* surprising to the model. The multilingual model may have seen more real, messy Banglish online than the polished style chatbots produce. The ChatGPT texts were also shorter (a median of 9 words, against 14), and short texts tend to get higher perplexity, so length probably adds to the gap.

Breaking the results down by generator shows the same pattern:

Generator	ChatGPT-RoBERTa	GPT-2 detector	Perplexity
Gemini, passed as human	100%	38%	42%
ChatGPT, passed as human	100%	44%	76%

Overall, **26 of the 200 texts fooled all three detectors at once.**

Where the detectors went wrong

Here are four real examples from the GPT-2 detector, which made the most confident mistakes. The "AI score" is the detector's probability that the text was written by AI.

Text	Actually written by	AI score
"onek onek valo ....onek comfortable...but Ami life a first time nilam tai vul size order kore felchi tai akto tight hocche..."	Human (2022 review)	**99.8%**
"Alhamdulillah original ta paici Thanks a lot 😊❤️ *Rider er behavior o valo cilo* ❤️*"*	Human (2022 review)	**99.6%**
"Bhaia er theke taka dhar niye shopping e gelam, ekhon dekhchi amar budget er theke double lagche."	Gemini	**0.8%**
"New earbuds er sound clarity khub impressive, call quality o bhalo."	ChatGPT	**0.7%**

The two human reviews are as human as text gets. They have repeated words, trailing dots, emoji, and a personal story about ordering the wrong size. The detector was almost certain both came from AI. Meanwhile, a clean, slightly generic ChatGPT sentence about earbuds scored under 1%.

I can only guess at the reasons. This detector was built in 2019 to spot GPT-2 output, which was often repetitive and a bit strange, so Banglish's repeated words and unusual spellings may look "machine-like" to it. Whatever the cause, its verdicts on Banglish had little to do with who actually wrote the text.

Limitations

This was a small test, so it isn't a benchmark:

- It used 200 samples, three detectors and one domain (product reviews).
- The ChatGPT texts were shorter than the others, and length affects perplexity and possibly the classifiers too.
- The human reviews are from 2018 to 2022, while the AI texts were generated in 2026, so writing style may have drifted between them.
- The perplexity detector used a simple median threshold. A tuned threshold might do somewhat better, but it couldn't fix a signal that points the wrong way.
- I didn't test commercial detectors such as GPTZero.

The results show a clear pattern worth investigating, but they aren't a final verdict on any tool.

4. What could work better

Multilingual models trained on Banglish. Models like XLM-R and mBERT have seen many languages during pre-training. Fine-tuning one on labelled Banglish examples, human and AI, should give it a much better sense of what "normal" Banglish looks like.

Banglish-specific features. Some signals come from the language itself: how often a text switches between Bangla and English, how consistent its spelling is, and how varied its vocabulary is. My examples hint that chatbot Banglish is cleaner and more standardised than what real customers write. Simple features like these can be combined with a neural model, and they're easy to interpret.

Explainability. A detector that only outputs "99.8% AI" gives the person being judged no way to question it. Look again at the first review in the table: a real customer would have no way to prove they wrote it. Methods like **SHAP** and **Integrated Gradients** can highlight which words or features pushed the model toward its decision. That gives teachers and moderators a reason they can check, and it gives researchers a way to catch a detector that has learned the wrong thing, such as "unusual spelling means AI."

This is the direction I'm taking in my technical project, where I'm building an explainable detector for AI-generated Banglish text.

5. Conclusion

- **Detectors learn what "normal" text looks like, mostly from standard English.**
- **Banglish breaks that assumption.** On 200 Banglish reviews, three detectors scored between 41% and 50%, no better than guessing. One missed every AI text, and another flagged 71% of real human reviews as AI.
- **Better data, multilingual models and explainability** are the most promising ways forward.

If you write in Banglish, your everyday writing style is exactly the kind of data these tools are missing. Until detectors are tested on it, a score from one of them shouldn't be used to judge a Banglish writer. If you'd like to contribute samples or try the notebook, [https://colab.research.google.com/drive/1A_5cKP-HulXIOz3C2_SC3QrxM52M-dKa#scrollTo=dXp9tNyAcbGB].

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